

Shreyash Wetal

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PROFESSIONAL SUMMARY

Postgraduate Computer Science student at the University of Sydney specialising in Artificial Intelligence, Data Science, Generative AI, and applied innovation systems. Experienced in using LLMs, RAG pipelines, prompt engineering, Python, Excel, dashboards, and automation workflows to support research, ideation, analytics, and decision-making. Strong hands-on background in building AI-enabled tools for content generation, insight extraction, evaluation, data processing, and practical business problem-solving. Interested in applying AI to product innovation, consumer insights, sports nutrition, health and wellness, concept testing, trend scanning, and market-driven product development.

TECHNICAL SKILLS

- AI & GenAI Tools:** ChatGPT, Gemini, OpenAI API, LangChain, prompt engineering, RAG, embeddings, vector databases, LLM evaluation, AI-assisted research, content generation
- Innovation & Product Development:** AI-enabled ideation, concept generation, product claim research, ingredient trend scanning, competitor analysis, idea scoring, product testing support
- Data & Consumer Insights:** Python, SQL, Excel, Pandas, data cleaning, structured/unstructured data analysis, consumer insight analysis, market research, dashboard reporting
- Machine Learning:** Supervised and unsupervised learning, deep learning, CNNs, transfer learning, model optimization, hyperparameter tuning, evaluation metrics
- Visualization & Dashboards:** Excel dashboards, Python reporting, data summaries, visual outputs, presentation-ready insights, analytics templates
- Cloud & Deployment:** Google Cloud Platform, Firebase, REST APIs, CI/CD, scalable inference systems, Flask, Django, Git/GitHub
- Computer Vision & NLP:** OpenCV, YOLOv8, object detection, image classification, semantic segmentation, transformers, NLP pipelines, conversational AI
- Certifications:** DeepLearning.AI TensorFlow Developer Professional Certificate; Deep Learning Specialization; LangChain LLM Applications; JP Morgan Quantitative Modelling

WORK EXPERIENCE

Research Student (Artificial Intelligence) | University of Sydney

March 2025 – Present

- Designed and trained deep learning models for medical image analysis, improving Dice score by 12% through preprocessing, augmentation, and model tuning.
- Implemented cross-validation, evaluation pipelines, and statistical performance tracking using IoU, F1-score, precision-recall, and ROC metrics.
- Benchmarked U-Net and DeepLabV3+ architectures, optimizing model robustness, inference efficiency, and reproducible experimentation.
- Applied Python-based analysis workflows to interpret model performance, compare experiments, and generate structured research insights.

Artificial Intelligence Engineer (Team Lead), The Modern Group | Sydney

June 2025 – Dec 2025

- Built LLM-based systems using RAG pipelines, vector retrieval, prompt engineering, and long-term context retrieval, improving retrieval accuracy by 30%.
- Designed AI-driven decision support workflows for research, automation, content generation, and data-driven recommendations, improving contextual relevance by 25%.
- Created reusable AI workflow templates and prompt frameworks to support faster ideation, structured analysis, and business process automation.
- Built scalable real-time inference APIs integrated with Firebase and GCP, reducing deployment cycle time by 40%.
- Designed ETL pipelines, data models, and analytics workflows for 100K+ records, supporting reporting, trend identification, and operational insights.

Associate Prompt Engineer, NVIDIA | Santa Clara, CA

Oct 2024 – Dec 2024

- Engineered enterprise-grade prompt frameworks improving LLM response precision by 22%.
- Conducted A/B testing across transformer-based models to compare accuracy, clarity, consistency, and output usefulness.
- Developed automated prompt evaluation pipelines to standardize AI output quality assessment and NLP benchmarking.
- Applied feedback-based refinement strategies to improve response reliability, relevance, and generation quality.

PROJECTS

AI-Powered Product Innovation & Consumer Insights Framework

- Built a GenAI workflow using ChatGPT/Gemini to generate product concepts, consumer personas, claims, positioning ideas, and market research summaries.
- Designed an idea scoring framework to evaluate concepts based on feasibility, consumer relevance, differentiation, evidence strength, and commercial potential.
- Used Python and Excel to clean data, summarize insights, compare trends, and create dashboard-style outputs for decision-making.
- Developed presentation-ready concept summaries and visual outputs to support AI-assisted ideation, concept testing, and innovation pipeline planning.

Medicinal Plant Identification with AI Chatbot

- Developed a CNN-based image classification system for medicinal plant recognition, connecting AI with natural health and wellness use cases.
- Integrated a LangChain-powered chatbot to retrieve plant knowledge and provide real-time explanatory responses.
- Designed an end-to-end AI pipeline combining computer vision inference, NLP retrieval, and user-facing conversational AI.

EyeDentify – Deep Learning Biometric Identification (AJCAI 2025, Springer)

- Developed a deep learning biometric authentication system using spatio-temporal blink modeling and feature extraction.
- Implemented MobileNetV2 achieving 99.04% accuracy on NTHU-DDD and 94.78% accuracy on UTA-RLDD datasets.
- Published peer-reviewed research at AJCAI 2025, demonstrating applied AI research, experimentation, and technical communication.

ACHIEVEMENTS

- Published peer-reviewed research at AJCAI 2025 (Springer) and ICSC 2024.
- Published “Intelligent Auto Traffic Signal Controller for Emergency Vehicles” in International Journal of Science and Applied Research.
- FIAT India Academic Excellence Scholarship Recipient from 2020–2024.
- Top 20 National Finalist – RoboKidz International Robotics Competition.

EDUCATION

University of Sydney

Master of Computer Science – Artificial Intelligence and Data Science

2025–2027

WAM: 75

Pimpri Chinchwad College of Engineering

Bachelor of Technology in Information Technology

2020–2024

CGPA: 8.57/10